

AWS DevOps Engineer Professional

In-Scope Services Reference Notes

What each service does, when to use it, and how it differs from similar services

ANALYTICS

Amazon Athena

Property	Detail
What it is	Interactive query service — run SQL against data in S3 without loading it anywhere
Cost model	Pay per TB of data scanned. Use columnar formats (Parquet, ORC) to reduce scan size and cost
When to use	Ad-hoc, infrequent analysis of S3 data (logs, access logs, CloudTrail). Cost-effective at any scale
NOT for	Real-time queries, continuous ingestion, OLTP workloads
vs Redshift	Athena = serverless, no cluster, pay per query. Redshift = provisioned cluster, expensive, better for continuous heavy analytics
vs OpenSearch	Athena = SQL on S3 files. OpenSearch = real-time log search and dashboards
Exam pattern	S3 access logs / CloudTrail logs / ALB logs analysis → always Athena for cost-effective solution

Amazon EMR

Property	Detail
What it is	Managed cluster platform for running big data frameworks at scale
When to use	Large-scale ETL, ML training jobs, complex data transformations requiring distributed compute
Cost model	Provisioned cluster — pay while running even if idle. Expensive for occasional use
vs Athena	EMR = complex distributed processing, long-running jobs. Athena = simple SQL, ad-hoc, serverless
Exam pattern	Usually a distractor — if question asks cost-effective log analysis, Athena wins over EMR

Amazon Kinesis Data Firehose

Property	Detail
What it is	Fully managed delivery stream. Ingests → optionally transforms (Lambda) → delivers to destination
Destinations	S3, Redshift, OpenSearch, Splunk, HTTP endpoint
Buffering	Buffers data before delivery (configurable by size or time interval). NOT real-time — near real-time
vs Kinesis Data Streams	Firehose = managed delivery, no consumers to write. Streams = real-time, you write consumers (KCL/Lambda)
Transformation	Optional Lambda function to transform records before delivery
Exam pattern	CloudWatch Logs → Firehose → S3 for log archiving. WAF logging → Firehose (name must start aws-waf-logs-)

Amazon Kinesis Data Streams

Property	Detail
What it is	Real-time data stream. Producers write records, consumers read and process them
Retention	Default 24 hours, up to 365 days (vs Firehose which delivers and doesn't retain)
Shards	Throughput unit — 1 shard = 1MB/s in, 2MB/s out. Max 2 consumers per shard (5 with Enhanced Fan-Out)
KCL metric: MillisBehindLatest	How far behind the consumer is. Scale ASG based on this metric when consumers lag
vs Firehose	Streams = real-time processing, you manage consumers. Firehose = managed delivery to storage/analytics
vs SQS	Kinesis = ordered, replay-able, multiple consumers, time-series. SQS = queue, message deleted after consumption
DynamoDB Streams fan-out	DynamoDB Streams → Kinesis → multiple Lambda consumers (avoids 2-consumer limit on DynamoDB Streams)

Amazon OpenSearch Service

Property	Detail
What it is	Managed OpenSearch (formerly Elasticsearch) cluster for search and analytics
Use cases	Real-time log analytics, full-text search, security event correlation, clickstream analysis
Ingestion patterns	Kinesis Firehose → OpenSearch (native). CloudWatch Logs → Subscription → Lambda → OpenSearch
vs Athena	OpenSearch = real-time search on continuously ingested data. Athena = ad-hoc SQL on S3 files
vs CloudWatch Logs Insights	OpenSearch = complex analytics, dashboards, cross-source. CW Insights = simpler, within CloudWatch only
Exam pattern	Usually overkill — if question says cost-effective or minimal effort, Athena wins. OpenSearch for real-time dashboards

Amazon QuickSight

Property	Detail
What it is	Serverless BI service for creating visualizations and dashboards from AWS data sources
Common pattern	S3 data → Athena queries → QuickSight dashboard
Exam role	Usually the visualization layer in a data pipeline — appears as the final step after Athena or Firehose

APPLICATION INTEGRATION

Amazon AppFlow

Property	Detail
What it is	Fully managed flow service to transfer data between Salesforce, ServiceNow, Slack etc. and AWS services
Destinations	S3, Redshift, Snowflake, and other SaaS apps
Exam context	Rarely tested — appears when question involves SaaS-to-AWS data transfer without custom code

Amazon EventBridge

Property	Detail
What it is	Serverless event bus. Rules match incoming events and route to targets (Lambda, SQS, SNS, Step Functions etc.)
Event sources	AWS services (aws.codepipeline, aws.ec2, aws.health etc.), SaaS partners, custom events
Key feature	Content-based filtering in rules — filter on specific field values within event JSON
vs SNS	EventBridge = event routing with filtering. SNS = pub/sub notification fan-out
Config → EventBridge	Use for rule-specific Config alerts. Direct Config → SNS streams ALL events (no filtering)
Trusted Advisor	Publishes to EventBridge (source: aws.trustedadvisor) — NOT directly to SNS
Health events	AWS_RISK_CREDENTIALS_EXPOSED comes from aws.health source in EventBridge

COMPUTE

AWS App Runner

Property	Detail
What it is	Managed service that builds, deploys, and runs web apps and APIs automatically
You provide	Source code (GitHub) or container image (ECR)
It handles	Build, deploy, load balancing, auto-scaling, TLS, health checks
Workloads	Stateless web services and APIs only
vs Elastic Beanstalk	App Runner = even simpler, less control. EB = more control, deployment strategies, .ebextensions
vs ECS/Fargate	App Runner = no container/cluster knowledge needed. ECS = full control, complex orchestration
Exam context	Usually a distractor — DevOps Pro tests detailed control. App Runner abstracts too much away

EC2 Image Builder

Property	Detail
What it is	Managed service to automate creation, maintenance, and distribution of AMIs and container images
Pipeline stages	Source base image → Build (install software) → Test (run tests) → Distribute (copy to regions/accounts)
vs SSM Automation	Image Builder = purpose-built for AMI creation pipelines. SSM Automation = general automation, can create AMIs but more complex
vs manual AMI creation	Image Builder automates the entire process on a schedule — no manual steps
Exam pattern	Automated weekly golden AMI creation → Image Builder or SSM Automation document

AWS Serverless Application Repository (SAR)

Property	Detail
What it is	Managed repository for discovering, deploying, and publishing serverless (Lambda-based) applications
Format	Applications packaged as AWS SAM templates
Visibility	Private (your account), shared (specific accounts), or public (everyone)
vs Service Catalog	SAR = serverless/Lambda apps only. Service Catalog = any AWS infrastructure via CloudFormation
Exam trap	SAR appears as distractor for general infrastructure sharing questions — it is Lambda-specific only

CONTAINERS

Container Services Overview

Service	What it is	Exam context
Amazon ECR	Private Docker registry. Regional service. Cross-region replication available	Always the image source for ECS/EKS. Use sha256 digest not :latest
Amazon ECS	Container orchestration on EC2 or Fargate. Two launch types	Blue/Green via CodeDeploy. Task definitions. awslogs driver for CloudWatch
Amazon EKS	Managed Kubernetes. More complex than ECS but full K8s API	aws-auth ConfigMap required for cross-account. No CodeDeploy native integration
AWS Fargate	Serverless compute for ECS and EKS. No EC2 instances to manage	ECS on Fargate supports CodeDeploy Blue/Green. EKS on Fargate does not
AWS Copilot	CLI tool to deploy containerised apps to ECS/Fargate. Abstracts infrastructure	Rarely tested — simplification tool for developers
AWS App2Container	Migrates existing .NET and Java apps into containers automatically	Migration tool — appears in modernisation scenarios
EKS Distro	AWS distribution of Kubernetes for running on-premises or in other clouds	Hybrid/on-premises K8s scenarios
ROSA	Red Hat OpenShift managed on AWS — fully managed OpenShift platform	Rarely tested — enterprise OpenShift scenarios

ECS Key Concepts

Concept	Detail
executionRoleArn	Used by ECS AGENT to pull images from ECR and write logs to CloudWatch. Not your app code
taskRoleArn	Used by APPLICATION CODE inside the container to call S3, DynamoDB, SQS etc.
awslogs driver	EC2 instance (not task) writes to CloudWatch. IAM role on EC2 instance, not task role
Image digest	Use @sha256:abc... in task definition — forces fresh pull, enables SHA tracking in CloudWatch Events
Blue/Green via CodeDeploy	Supported for ECS on EC2 and ECS on Fargate. NOT supported for EKS
Fargate CPU/memory	Must use valid combinations — invalid combinations fail task definition registration

DATABASE

Database Services Quick Reference

Service	Type	Key exam points
Amazon Aurora	MySQL/PostgreSQL compatible relational	Global Database for cross-region. IAM authentication (no credentials). Blue/Green deployments for major version upgrades
Aurora Serverless v2	Auto-scaling serverless Aurora	Scales to zero. Good for intermittent workloads. Zero-downtime minor upgrades
Amazon RDS	Managed relational DB	Multi-AZ for HA. Read Replicas for scale. Major upgrades need Read Replica promotion. EngineVersion not DBEngineVersion in CFN
Amazon DynamoDB	NoSQL key-value and document	DynamoDB Streams for item-level changes. DAX for microsecond reads. No credentials — IAM only. Global Tables for multi-region
Amazon ElastiCache	In-memory cache	Redis = persistence, complex data. Memcached = simple cache. Cannot cache DynamoDB (use DAX instead)
Amazon MemoryDB for Redis	Durable Redis-compatible in-memory DB	Unlike ElastiCache Redis, MemoryDB is a primary database — fully durable with Multi-AZ
Amazon Redshift	Data warehouse	Complex analytics at scale. Redshift Spectrum queries S3 without loading. Expensive — overkill for simple log analysis
Amazon DocumentDB	MongoDB-compatible document DB	Secrets Manager supports native rotation. Not DynamoDB — different use case
AWS DMS	Database Migration Service	Migrate databases to AWS with minimal downtime. Supports homogeneous and heterogeneous migrations

DynamoDB Streams vs CloudTrail

Event	Streams captures?	CloudTrail captures?
PutItem / UpdateItem / DeleteItem (item operations)	YES	YES (as API call)
DeleteTable (table-level API)	NO	YES — use CWE rule on CloudTrail
Max concurrent consumers per shard	2 (throttles beyond this)	N/A
Fan-out solution when > 2 consumers needed	DynamoDB Streams → SNS → multiple Lambdas	N/A

DEVELOPER TOOLS

Developer Tools Quick Reference

Service	What it does	Key exam points
AWS CodeArtifact	Managed artifact repository for npm, Maven, pip, NuGet packages	Proxy public repos (npmjs, PyPI). Share packages across accounts. Reduces external dependency on internet
AWS CodeBuild	Managed CI build service — compile, test, package	Max 8hrs. Cache types: Local (fast, same host) vs S3 (persistent across hosts). Docker layer cache for container builds
AWS CodeDeploy	Deployment automation to EC2, Lambda, ECS, on-premises	Canary/Linear = Lambda+ECS only. EC2 canary = 2 deployment groups. ValidateService = last lifecycle hook
Amazon CodeGuru	ML-powered code reviews and profiling	Reviewer: security issues, hardcoded secrets in PRs. Profiler: runtime performance bottlenecks
AWS CodePipeline	CI/CD orchestration — chains source, build, deploy	runOrder same value = parallel. Cannot invoke another pipeline directly. Cross-region via S3 copy
AWS CodeStar	Unified UI wrapping CodeCommit/Build/Deploy/Pipeline	NOT a stage action type. Distractor in pipeline questions
AWS FIS	Fault Injection Simulator — controlled chaos engineering	Run experiments to test resilience. Inject failures into EC2, ECS, RDS, networking
AWS X-Ray	Distributed tracing — trace requests across microservices	Service map shows request flow. Identify bottlenecks and errors across Lambda, EC2, ECS, API Gateway
AWS CDK	Define infrastructure as code using Python, TypeScript, Java etc.	Compiles to CloudFormation. Higher level abstraction than raw CFN templates

CodeBuild Cache — Quick Reference

Scenario	Cache Type	Cache Mode
Large Docker container, one build at a time, frequent	Local (faster, same host likely reused)	Docker layer
Large Docker container, concurrent builds	S3 (guaranteed cross-host persistence)	Docker layer
Slow dependency install (npm/pip/maven)	S3 or Local	Custom (specify paths)
Slow source checkout	S3 or Local	Source

MANAGEMENT AND GOVERNANCE

Management Services Quick Reference

Service	What it does	Key exam points
AWS CloudFormation	IaC — provision and manage AWS resources via templates	StackSets for multi-account/region. SSM parameter type for dynamic values. Custom Resource for unsupported resources. InsufficientCapabilitiesException = IAM capability needed
AWS CloudTrail	API call logging — who did what, when, from where	validate-logs for integrity. Use CWE rule on CloudTrail for specific API alerts. NOT for item-level DB changes
Amazon CloudWatch	Metrics, alarms, dashboards for AWS resources	Alarms → SNS (not Lambda directly). Metric Filter → metric → alarm. 15 month max retention
Amazon CloudWatch Logs	Centralised log storage and analysis	Metric Filter for alarms. Subscription Filter for real-time delivery to Lambda/Kinesis/Firehose
AWS Compute Optimizer	ML recommendations for right-sizing EC2, Lambda, ECS	Cost optimisation — recommends instance types based on actual usage patterns
AWS Config	Resource configuration tracking and compliance rules	Managed rules (no code). Custom rules (Lambda). Aggregator for multi-account. EventBridge not SNS for specific rule alerts
AWS Control Tower	Landing zone for multi-account AWS setup with guardrails	Sits on top of Organizations. Guardrails = SCPs + Config rules. Account Factory for new accounts
AWS Health	Personalised AWS service health and account-specific events	AWS_RISK_CREDENTIALS_EXPOSED comes from Health. EventBridge source: aws.health
AWS License Manager	Track and manage software licenses (per-socket, per-core)	Dedicated Hosts for socket-based licensing. License Manager enforces rules
Amazon Managed Grafana	Managed Grafana dashboards — visualise metrics from multiple sources	Works with CloudWatch, Prometheus, X-Ray, OpenSearch as data sources
Amazon Managed Prometheus	Managed Prometheus monitoring for container workloads	Pairs with Managed Grafana. For EKS/ECS container metrics at scale
AWS OpsWorks	Managed Chef/Puppet — EOL May 2024. Replaced by SSM	Configure lifecycle event fires on ALL instances when any joins/leaves. Used in Cassandra cluster exam questions
AWS Organizations	Multi-account management — SCPs, consolidated billing, OUs	SCPs = maximum permissions guardrail (don't grant, only limit). StackSets with service-managed permissions auto-deploy to new accounts
AWS Proton	Managed deployment service for container and serverless apps	Platform teams define templates. Dev teams self-serve. Rarely tested
AWS Resilience Hub	Assess and improve application resilience against disruptions	Define RTO/RPO targets. Resiliency score. Recommendations for improvement
AWS Service Catalog	Portfolio of approved CloudFormation products for end users	Beginner users launch compliant resources without knowing AWS. TagOptions for tagging enforcement
AWS Systems Manager	Operations hub — patch, run commands, manage config, store params	Run Command (ad-hoc), Automation (workflows), Inventory (software list), Patch Manager, Parameter Store

Service	What it does	Key exam points
AWS Trusted Advisor	Best practice checks — cost, performance, security, fault tolerance	Publishes to EventBridge (NOT SNS directly). Low utilisation EC2 check for cost savings

NETWORKING AND CONTENT DELIVERY

Networking Services Quick Reference

Service	What it does	Key exam points
Amazon API Gateway	Managed API service — REST, HTTP, WebSocket APIs	29 sec max integration timeout. Canary deployments on stages. Static mapping templates (not conditional defaults). Cognito for end-user auth, IAM for internal
AWS Client VPN	Managed VPN for users to securely access AWS and on-premises	Client-based VPN — individual user connections
Amazon CloudFront	CDN — global content delivery with edge caching	S3 + CloudFront = standard static content pattern. Lambda@Edge for custom logic at edge. OAC/OAI for S3 private access
Elastic Load Balancing	Distribute traffic across targets	ALB = HTTP/HTTPS, path/host routing. NLB = TCP/UDP, static IP. ALB health check = HTTP status code. Access logs at ALB level not target group
AWS PrivateLink	Private connectivity to AWS services without internet gateway	Service endpoint stays in VPC. No public IP needed. Used for VPC-to-service connectivity
Amazon Route 53	DNS service — routing, health checks, domain registration	Health checks are DNS-level — cannot route between instances behind ALB. Weighted/latency/failover routing policies
AWS Site-to-Site VPN	IPSec VPN between on-premises and AWS VPC	Network-level encryption. Lower cost than Direct Connect. Higher latency
AWS Transit Gateway	Hub for connecting multiple VPCs and on-premises networks	Replaces complex VPC peering meshes. Central hub routing
Amazon VPC	Isolated virtual network in AWS	Subnets, route tables, security groups, NACLs. VPC peering for cross-account/cross-region connectivity

SECURITY, IDENTITY, AND COMPLIANCE

Security Services Quick Reference

Service	What it does	Key exam points
AWS ACM	Managed TLS/SSL certificates	Free for AWS services (ALB, CloudFront). Auto-renewal. Cannot export private key for ACM-managed certs
AWS CloudHSM	Dedicated hardware security module — you control keys	FIPS 140-2 Level 3. Full key control. More expensive than KMS. For strict compliance requirements
Amazon Cognito	User authentication for web and mobile apps	User Pools = sign-up/sign-in. Identity Pools = AWS credentials for authenticated users. Use for public API auth (not IAM)
Amazon Detective	Investigate security findings — visualise and analyse suspicious activity	Works with GuardDuty findings. Root cause analysis of security events
AWS Directory Service	Managed Microsoft Active Directory in AWS	Extends on-premises AD to AWS. Use for Windows workloads needing AD
Amazon GuardDuty	Threat detection — malicious activity, anomalous behaviour	Analyses CloudTrail, VPC Flow Logs, DNS logs. NOT for compliance rules or vulnerability scanning
AWS IAM Identity Center	SSO for multiple AWS accounts and business apps	Replaces AWS SSO. Central identity management across org
AWS IAM	Access control — users, roles, policies, permissions	Explicit Deny always wins. PassRole needed to hand role to service. Trust policy on role being assumed
Amazon Inspector	Vulnerability scanning — EC2, ECR, Lambda	Scans for CVEs. Needs running EC2 instance (cannot scan AMI directly). NOT for config compliance (use Config)
AWS KMS	Managed encryption key service	Per-API-call charge. At 10k+ objects/sec use SSE-S3 not SSE-KMS (throttle risk). Automatic rotation available
Amazon Macie	PII and sensitive data discovery in S3	ML-based. Detects PII, credit cards, credentials in S3 objects. NOT GuardDuty (threats) or Inspector (vulnerabilities)
AWS Network Firewall	Managed stateful network firewall for VPC	Layer 3-7 filtering. Intrusion prevention. For network-level threat protection
AWS RAM	Share AWS resources across accounts	Share subnets, Transit Gateway, Route 53 Resolver rules etc. within org without duplicating
AWS Secrets Manager	Store and rotate secrets — DB credentials, API keys	Native rotation for RDS, Aurora, Redshift, DocumentDB. NOT DynamoDB (IAM-based). Cross-region replication built in
AWS Security Hub	Centralised security findings from multiple services	Aggregates GuardDuty, Inspector, Macie, Config findings. Single pane of glass for security posture
AWS STS	Temporary security credentials	AssumeRole for cross-account. AssumeRoleWithSAML for SAML federation. AssumeRoleWithWebIdentity for OIDC
AWS Shield	DDoS protection	Standard = automatic, free for all. Advanced = enhanced protection, 24/7 DRT, financial protection
AWS WAF	Web Application Firewall — filter HTTP requests	Rules on ALB, CloudFront, API Gateway. Logs must go to S3/Firehose/CWL with aws-waf-logs- prefix

Security Service	Use for	NOT for
GuardDuty	Threat detection — malicious activity, compromised instances, unusual API calls	Compliance rules, vulnerability scanning, PII detection
Inspector	Vulnerability scanning — CVEs in OS/software on EC2, ECR, Lambda	Config compliance, threat detection, PII

Security Service	Use for	NOT for
Macie	Sensitive data discovery — PII, credentials in S3	Threat detection, vulnerability scanning, network monitoring
Config	Resource configuration compliance over time	Threat detection, vulnerability scanning, PII
CloudTrail	API call audit trail — who did what	Real-time threat detection, item-level DB changes

SERVERLESS

Serverless Services Quick Reference

Service	What it does	Key exam points
AWS Lambda	Run code without servers — event-driven, auto-scaling	Max 15 min timeout. Max 10GB memory. Triggers: S3, DynamoDB Streams, Kinesis, SQS, SNS, API GW, EventBridge, CodePipeline
AWS SAM	Framework for defining serverless apps — extends CloudFormation	SAM template compiles to CloudFormation. Local testing with sam local. Deployed via SAR for sharing
Amazon SNS	Pub/sub messaging — one topic, many subscribers	Fan-out pattern. Cannot filter by Config rule (use EventBridge). Cannot directly trigger from CloudWatch Alarm → Lambda (need SNS in between)
Amazon SQS	Message queue — decouple producers and consumers	Standard = at-least-once, best-effort ordering. FIFO = exactly-once, ordered. DLQ for failed messages. 256KB max message size
AWS Step Functions	Orchestrate multi-step workflows with state management	Standard = 1 year max, full audit trail. Express = 5 min max, high volume. Use for complex workflows with retries/error handling. NOT for simple Lambda tasks

SNS vs SQS	SNS	SQS
Pattern	Pub/sub — push to all subscribers	Queue — pull by one consumer
Message retention	Not retained — delivered or lost	Up to 14 days
Consumers	Multiple simultaneous (fan-out)	One consumer per message
Use when	Notify multiple systems of same event	Decouple and buffer work between producer and consumer

STORAGE

Storage Services Quick Reference

Service	What it does	Key exam points
AWS Backup	Centralised backup across AWS services	Supports EFS, DynamoDB, RDS, EC2, EBS, S3. Cross-region and cross-account backup. Vault Lock for WORM compliance
Amazon EBS	Block storage attached to EC2 instances	AZ-locked — cannot attach across AZs. Snapshot to copy cross-region. For stateful services (Kafka, Cassandra) use one ASG per AZ
AWS Elastic Disaster Recovery	Continuous replication for fast failover to AWS	RPO measured in seconds. RTO in minutes. For on-premises DR to AWS
Amazon EFS	Managed NFS file system — shared across multiple EC2 instances	Regional service (not cross-region). Multi-AZ native. Cross-account via VPC peering + resource policy. EFS replication for DR (read-only replica)
Amazon FSx for Lustre	High-performance parallel file system for HPC and ML	Integrates with S3 — can use S3 as data repository
Amazon FSx for NetApp ONTAP	Managed NetApp ONTAP — enterprise NAS features	Multi-protocol (NFS, SMB, iSCSI). Multi-AZ. For enterprise storage workloads
Amazon FSx for OpenZFS	Managed OpenZFS file system	ZFS features — snapshots, cloning, compression. For workloads needing ZFS
Amazon FSx for Windows	Managed Windows File Server — SMB protocol, AD integration	Windows workloads needing native Windows file sharing
Amazon S3	Object storage — unlimited scale, 11 nines durability	SSE-S3 for high throughput (no KMS throttle). CRR for cross-region replication. Bucket policy with aws:SecureTransport for HTTPS enforcement
Amazon S3 Glacier	Long-term archival storage — low cost, slow retrieval	Vault Lock for WORM (compliance). Retrieval: Expedited mins, Standard hrs, Bulk hrs-days
AWS Storage Gateway	Hybrid storage — connect on-premises to AWS storage	File Gateway (S3), Volume Gateway (EBS), Tape Gateway (Glacier). For on-premises apps needing cloud storage

Storage Scope & Cross-Region Capabilities

Service	Scope	Cross-region native?	Cross-account?
S3	Global namespace, regional storage	YES — CRR (Cross-Region Replication)	YES — bucket policy
EBS	AZ-locked	NO — snapshot copy only	NO — snapshot share
EFS	Regional (multi-AZ)	NO — EFS replication (read-only)	YES — VPC peering + policy
ECR	Regional	YES — cross-region replication	YES — resource policy
Secrets Manager	Regional	YES — built-in replication	YES — resource policy
SSM Parameter Store	Regional	NO — manual copy needed	YES — resource policy

Key Exam Decision Rules

If the question mentions...	Think...
Cost-effective log analysis on S3	Athena — serverless, pay per query
Real-time log dashboards	OpenSearch + OpenSearch Dashboards
Long-running task (> 15 min)	CodeBuild or Step Functions — never Lambda
Alert on specific Config rule violation	Config → EventBridge → SNS (not Config → SNS direct)
Specific API call notification (DeleteTable etc.)	CloudTrail → CloudWatch Events → SNS (not DynamoDB Streams)
DynamoDB access (no credentials)	IAM role only — Secrets Manager does not support DynamoDB
Cross-account trust relationship	Trust policy on the DESTINATION role — caller gets sts:AssumeRole
EKS cross-account access	IAM role + aws-auth ConfigMap — both required
Prevent direct pushes to master in CodeCommit	IAM Deny policy with codecommit:References condition — not repository policy (doesn't exist)
Canary deployment on EC2	Two CodeDeploy deployment groups + manual approval — not CodeDeploy canary config (Lambda/ECS only)
High throughput S3 encryption (10k+ objects/sec)	SSE-S3 not SSE-KMS — KMS throttles at high request rates
Scan for PII in S3	Amazon Macie
Scan for vulnerabilities / CVEs	Amazon Inspector
Detect malicious activity / threats	Amazon GuardDuty
Resource configuration compliance over time	AWS Config
Beginner users need to create compliant resources	AWS Service Catalog — not CloudFormation directly
Multi-account IAM role + CloudTrail enforcement	CloudFormation StackSets with service-managed permissions
Credentials exposed on GitHub alert	AWS Health event (aws.health) → EventBridge → Step Functions